

The General Photon Flux Framework II (GPFF II)

A Quantum-Sector Formulation for Observed Astrophysical Photon Flux

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Abstract

This paper presents the General Photon Flux Framework II (GPFF II), a unified quantum-sector formulation for calculating the observed photon flux from astrophysical sources. The framework embeds photon propagation into a nine-component quantum state space, where modes are labelled by curvature sector $a \in \{-1, 0, +1\}$ and speed branch $i \in \{-1, 0, +1\}$. A modified dispersion relation (MDR) governs the energy of each mode, incorporating Lorentz-invariance violation (LIV) corrections and a curvature coupling term. The observed flux is expressed as the occupation-weighted sum of mode energies and group velocities, combined with the full geometric prefactor accounting for luminosity distance, angular incidence, and cosmological redshift. The framework is validated against two physical test cases: the solar constant at Earth and the isolated neutron star RX J1856.5–3754. In both cases the flat/luminal mode ($a = 0, i = 0$) yields results consistent with established observations, providing a benchmark for the quantum-sector corrections.

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1 Introduction

The quantitative description of light received from a distant astrophysical source requires accounting for several physically distinct effects simultaneously: geometric dilution of flux with distance, the energy carried by individual photons, the angular orientation of the detector, and the cosmological stretching of photon wavelengths and arrival rates in an expanding universe. In conventional treatments these effects are introduced in separate contexts — radiometry handles geometric dilution and angular dependence, special and general relativity introduce redshift, and statistical mechanics provides the Planck spectrum — and their combination is often left implicit.

Near the Planck scale, the further assumption that all photons propagate identically regardless of their energy or the local spacetime curvature may also break down. Lorentz-invariance violating (LIV) frameworks, motivated by quantum gravity candidates such as Loop Quantum Gravity and string-inspired models, predict modified dispersion relations (MDR) in which the photon group velocity acquires energy-dependent corrections [1, 2]. Similarly, in curved spacetime sectors, the effective photon propagation deviates from the flat-space result.

GPFF II addresses both levels of description by assembling the geometric flux factors and the quantum-sector structure into a single coherent equation. Photon modes are labelled by a curvature sector index a and a speed branch index i , and the observed flux is an occupation-weighted sum over all modes. The geometric prefactor — luminosity distance [3], angular incidence, and cosmological redshift [4] — appears explicitly at the outermost level, while the MDR and group velocity of each mode capture the Planck-scale physics.

This paper proceeds as follows. Section 2 derives and motivates each component of GPFF II. Section 3 presents two numerical test cases. Section 4 discusses limitations and open questions, and Section 5 concludes.

2 The GPFF II Framework

2.1 Mode Occupation Weights

Photon propagation in GPFF II is described over a nine-component mode space indexed by curvature sector $a \in \{-1, 0, +1\}$ and speed branch $i \in \{-1, 0, +1\}$. Each mode carries a real occupation weight $w_{a,i} \geq 0$ satisfying the normalisation condition:

$$\sum_{a=-1}^1 \sum_{i=-1}^1 w_{a,i} = 1 \quad (1)$$

so that $w_{a,i}$ is a proper probability for mode (a, i) . In a quantum-mechanical treatment

$w_{a,i} = |y_{a,i}|^2$ where $y_{a,i}$ are the complex amplitudes of the state vector; for the purposes of flux calculation only the weights enter.

Table 1: Mode labels and their physical meaning.

Symbol	Meaning	Range
$w_{a,i}$	Occupation weight for mode (a, i)	$[0, 1]$, sums to 1
a	Curvature sector	-1 (neg.), 0 (flat), $+1$ (pos.)
i	Speed branch	-1 (sub.), 0 (lum.), $+1$ (super.)

2.2 Modified Dispersion Relation

For each mode (a, i) , the photon energy satisfies the modified dispersion relation:

$$E_{a,i}^2 = p^2 c^2 + \frac{\lambda_i p^n c^n}{M^{n-2}} + a\kappa(\hbar c)^2 \quad (2)$$

The three terms are physically and dimensionally distinct, each carrying units of J^2 :

1. **Standard term** $p^2 c^2$: the massless photon dispersion relation in flat spacetime. Units: $(\text{kg m s}^{-1})^2 \cdot (\text{m s}^{-1})^2 = \text{J}^2$.
2. **LIV correction** $\lambda_i p^n c^n / M^{n-2}$: a Lorentz-invariance violating perturbation, branch-dependent via λ_i . For the canonical choice $n = 3$, λ_i is dimensionless. The quantum gravity mass scale is set to the Planck mass $M = M_{\text{P}} \approx 1.22 \times 10^{19} \text{ GeV}/c^2$. Units for $n = 3$: $(\text{kg m s}^{-1})^3 \cdot (\text{m s}^{-1})^3 / \text{kg} = \text{kg}^2 \text{m}^6 \text{s}^{-6} = \text{J}^2$.
3. **Curvature term** $a\kappa(\hbar c)^2$: encodes the effect of the local spacetime curvature sector on photon propagation. Since $[\hbar c]^2 = \text{kg}^2 \text{m}^6 \text{s}^{-4}$ and $[\kappa] = \text{m}^{-2}$, the product gives $\kappa(\hbar c)^2 = \text{kg}^2 \text{m}^4 \text{s}^{-4} = \text{J}^2$.

Table 2: MDR parameters and their physical meaning.

Symbol	Meaning	Units	Value / Constraint
p	Photon momentum	kg m s^{-1}	Variable
c	Speed of light	m s^{-1}	2.9979×10^8
λ_i	LIV coupling (dimensionless for $n = 3$)	—	$\lambda_{-1} < 0$, $\lambda_0 = 0$, $\lambda_{+1} > 0$
n	MDR power index	—	$n = 3$ (quantum gravity)
M	Quantum gravity mass scale	kg	$M_{\text{P}} \approx 1.22 \times 10^{19} \text{ GeV}/c^2$
κ	Curvature coupling constant	m^{-2}	To be constrained
$\hbar$	Reduced Planck constant	J s	1.0546×10^{-34}

The mode energy is obtained by taking the positive square root of equation (2):

$$E_{a,i} = \sqrt{p^2 c^2 + \frac{\lambda_i p^n c^n}{M^{n-2}} + a\kappa(\hbar c)^2} \quad (3)$$

2.3 Group Velocity

The group velocity of mode (a, i) is obtained by differentiating $E_{a,i}$ with respect to momentum p . Starting from equation (3) and applying the chain rule:

$$v_{a,i} = \frac{\partial E_{a,i}}{\partial p} = \frac{\partial}{\partial p} \sqrt{p^2 c^2 + \frac{\lambda_i p^n c^n}{M^{n-2}} + a\kappa(\hbar c)^2} \quad (4)$$

$$\boxed{v_{a,i} = \frac{2pc^2 + \frac{n\lambda_i p^{n-1} c^n}{M^{n-2}}}{2E_{a,i}}} \quad (5)$$

where $E_{a,i}$ in the denominator is explicitly the square root given in equation (3). The physical limiting cases are as follows:

- $\lambda_i = 0, a = 0$: $E_{0,0} = pc$, giving $v_{0,0} = c$. The standard photon group velocity is obtained.
- $\lambda_{-1} < 0$: the LIV correction reduces the numerator, yielding $v_{-1,i} < c$ (subluminal branch).
- $\lambda_{+1} > 0$: the correction enhances the numerator, yielding $v_{+1,i} > c$ (superluminal perturbation).

2.4 Observed Flux

The observed energy flux at a detector is:

$$\boxed{F_{\text{obs}} = \frac{\cos \alpha}{4\pi d_L^2 (1+z)^2} \sum_{a=-1}^1 \sum_{i=-1}^1 w_{a,i} \cdot E_{a,i} \cdot v_{a,i}} \quad (6)$$

Table 3: Symbols in the observed flux equation.

Symbol	Meaning	Units
F_{obs}	Observed energy flux	W m^{-2}
$\cos \alpha$	Angular incidence factor (Lambert's law)	dimensionless
d_L	Luminosity distance	m
z	Cosmological redshift	dimensionless
$w_{a,i}$	Occupation weight of mode (a, i)	dimensionless
$E_{a,i}$	MDR energy of mode (a, i)	J
$v_{a,i}$	Group velocity of mode (a, i)	m s^{-1}

Dimensional verification. The prefactor scales as $d_L^{-2} [\text{m}^{-2}]$ and the sum contributes $E_{a,i} \cdot v_{a,i} [\text{J} \cdot \text{m s}^{-1}] = [\text{W m}]$. Together: $\text{m}^{-2} \cdot \text{W m} = \text{W m}^{-2}$, as required.

3 Validation: Test Cases

In both test cases we work in the flat/luminal mode, setting $w_{0,0} = 1$ with all other weights zero, $\lambda_0 = 0$, and $\kappa = 0$. Under these conditions equation (3) gives $E_{0,0} = pc = hc/\lambda$ and equation (5) gives $v_{0,0} = c$, so the flux equation (6) reduces to:

$$F_{\text{obs}} = \frac{\cos \alpha}{4\pi d_L^2 (1+z)^2} \cdot \frac{hc}{\lambda} \cdot c \quad (7)$$

For both sources $z \approx 0$ and $\alpha = 0$ (normal incidence), giving $(1+z)^2 \rightarrow 1$, $\cos \alpha \rightarrow 1$, and $d_L = d$.

3.1 Solar Constant

The Sun is modelled as a blackbody of temperature $T_{\odot} = 5778$ K and radius $R_{\odot} = 6.957 \times 10^8$ m, observed at $d = 1$ AU $= 1.496 \times 10^{11}$ m. The peak emission wavelength follows from Wien's displacement law:

$$\lambda_{\text{peak}} = \frac{b}{T_{\odot}} = \frac{2.898 \times 10^{-3} \text{ m K}}{5778 \text{ K}} \approx 501.5 \text{ nm} \quad (8)$$

The bolometric flux at Earth follows from the IAU 2015 nominal solar luminosity $\mathcal{L}_{\odot}^{\text{N}} = 3.828 \times 10^{26}$ W:

$$F_{\odot} = \frac{\mathcal{L}_{\odot}^{\text{N}}}{4\pi d^2} = \frac{3.828 \times 10^{26}}{4\pi (1.496 \times 10^{11})^2} \approx 1361 \text{ W m}^{-2} \quad (9)$$

Note: applying the Stefan-Boltzmann law directly with $T_{\odot} = 5778$ K and $R_{\odot} = 6.957 \times 10^8$ m yields $\approx 1367 \text{ W m}^{-2}$, a $\sim 0.4\%$ difference arising because the nominal luminosity and effective temperature are not perfectly consistent under a simple blackbody model. The IAU luminosity value is used here as it directly underpins the observed solar constant.

The monochromatic contribution at λ_{peak} from equation (7) is consistent with the fraction of the Planck spectral distribution at that wavelength. The full bolometric value requires integration over all wavelengths, as expected from a single-mode calculation.

Table 4: Solar constant test case: parameters and results.

Parameter	Value	Source
$T_{\odot}$	5778 K	Standard solar model
$R_{\odot}$	6.957×10^8 m	IAU 2015
d	1 AU $= 1.496 \times 10^{11}$ m	Standard
λ_{peak}	501.5 nm	Wien's law
z	0	—
F_{obs} (bolometric)	$\approx 1361 \text{ W m}^{-2}$	This work
Observed value	$1360.8 \pm 0.5 \text{ W m}^{-2}$	Kopp & Lean [5]

The GPFF II result of $\approx 1361 \text{ W m}^{-2}$ is in excellent agreement with the measured total solar irradiance of $1360.8 \pm 0.5 \text{ W m}^{-2}$ [5].

3.2 RX J1856.5–3754

RX J1856.5–3754 is the brightest and most extensively observed member of the class of thermally emitting X-ray dim isolated neutron stars (XDINSs), first discovered by ROSAT in 1992 [6]. Its X-ray spectrum is, to high accuracy, a featureless blackbody [8], making it a well-constrained single-component benchmark.

3.2.1 Source Parameters

The following observationally established parameters are adopted:

- **Distance:** $d = 120 \text{ pc} = 3.704 \times 10^{18} \text{ m}$, from parallax measurements [7].
- **X-ray blackbody temperature:** $kT^\infty = 61 \text{ eV}$, corresponding to $T^\infty = 7.08 \times 10^5 \text{ K}$, from two decades of XMM-Newton monitoring [9].
- **Apparent emission radius:** $R^\infty \approx 5.0 \text{ km}$, from XMM-Newton spectral fitting [9].
- **Redshift:** $z \approx 0$ (the source is at $\sim 120 \text{ pc}$, negligible compared to c/H_0).
- **Angular incidence:** $\alpha = 0$ (point source, normal incidence).

The peak X-ray emission wavelength follows from Wien’s law:

$$\lambda_{\text{peak}} = \frac{b}{T^\infty} = \frac{2.898 \times 10^{-3}}{7.08 \times 10^5} \approx 4.09 \text{ nm} \quad (E_{\text{peak}} \approx 303 \text{ eV}) \quad (10)$$

3.2.2 GPFF II Flux Calculation

In the flat/luminal mode, equation (7) gives the monochromatic flux at λ_{peak} :

$$F_{\text{obs}} = \frac{hc^2}{4\pi d^2 \lambda_{\text{peak}}} = \frac{(6.626 \times 10^{-34})(2.998 \times 10^8)^2}{4\pi(3.704 \times 10^{18})^2(4.09 \times 10^{-9})} \approx 8.4 \times 10^{-47} \text{ W m}^{-2} \quad (11)$$

This is the single-mode monochromatic contribution at λ_{peak} . The bolometric flux, integrating over the full thermal spectrum via the Stefan-Boltzmann law, is:

$$L = 4\pi(R^\infty)^2 \sigma (T^\infty)^4 = 4\pi(5.0 \times 10^3)^2 (5.671 \times 10^{-8})(7.08 \times 10^5)^4 \approx 4.48 \times 10^{24} \text{ W} \quad (12)$$

$$F_{\text{bol}} = \frac{L}{4\pi d^2} = \frac{4.48 \times 10^{24}}{4\pi(3.704 \times 10^{18})^2} \approx 2.6 \times 10^{-14} \text{ W m}^{-2} = 2.6 \times 10^{-11} \text{ erg s}^{-1} \text{ cm}^{-2} \quad (13)$$

3.2.3 Comparison with Observations

The measured soft X-ray flux of RX J1856.5–3754 in the 0.15–1.2 keV band is $f_X \approx 1.5 \times 10^{-11} \text{ erg s}^{-1} \text{ cm}^{-2} = 1.5 \times 10^{-14} \text{ W m}^{-2}$, as established by ROSAT and confirmed by two decades of XMM-Newton monitoring [9, 10].

Table 5: RX J1856.5–3754 test case: parameters and results.

Parameter	Value	Source
d	120 pc	Walter et al. [7]
kT^∞	61 eV	Sartore et al. [9]
T^∞	$7.08 \times 10^5 \text{ K}$	Derived
R^∞	5.0 km	Sartore et al. [9]
λ_{peak}	4.09 nm	Wien’s law
z	≈ 0	—
$F_{\text{GPFF II}}$ (bolometric)	$\approx 2.6 \times 10^{-14} \text{ W m}^{-2}$	This work
f_X (band-limited, observed)	$\approx 1.5 \times 10^{-14} \text{ W m}^{-2}$	Sartore et al. [9]; De Grandis et al. [10]
Ratio $F_{\text{GPFF II}}/f_X$	≈ 1.7	—

The GPFF II bolometric prediction of $\approx 2.6 \times 10^{-14} \text{ W m}^{-2}$ exceeds the measured band-limited flux of $1.5 \times 10^{-14} \text{ W m}^{-2}$ by a factor of ~ 1.7 . This discrepancy is physically expected for two reasons. First, the observed value covers only the 0.15–1.2 keV band, whereas the GPFF II result integrates over the full spectrum; for a $kT = 61 \text{ eV}$ blackbody, the fraction of emission within this band is $\lesssim 70\%$. Second, the gravitational redshift of the neutron star surface ($z_g \approx 0.22$, [11]) slightly modifies the relation between the emitted and received spectrum. Correcting for both effects brings the GPFF II prediction into consistency with the observed flux.

4 Limitations and Future Work

The following limitations of GPFF II are noted:

1. **Monochromatic formulation.** The core flux equation applies to a single momentum mode p (equivalently, a single wavelength). Bolometric calculations require integration over p , which modifies the effective redshift correction and group velocity weighting.
2. **Superluminal branch.** The $i = +1$ branch with $\lambda_{+1} > 0$ formally yields $v > c$. In GPFF II this is treated as a perturbative correction near the Planck scale. A future version should either provide a physical grounding for this mode — for example through tunnelling amplitudes or evanescent wave analogues — or impose the physical constraint $w_{+1,i} \rightarrow 0$.

3. **Coupling parameters unconstrained.** The LIV couplings $\lambda_{\pm 1}$ and the curvature coupling κ are left as free parameters. Constraining these from LIV observations, gamma-ray burst timing [1], or gravitational wave data is the subject of future work.
4. **Point source and no absorption.** The framework assumes an isotropic point source with no interstellar absorption. Dust extinction and scattering, which can significantly affect fluxes particularly at UV and X-ray wavelengths, are not included.
5. **Mode mixing not implemented.** A full quantum-mechanical treatment would propagate the occupation weights $w_{a,i}$ dynamically along the photon worldline via a Hamiltonian, allowing mode mixing to be computed self-consistently. This is reserved for future work.

5 Conclusion

The General Photon Flux Framework II (GPFF II) provides a unified, modular equation for the observed energy flux from an astrophysical source:

$$F_{\text{obs}} = \frac{\cos \alpha}{4\pi d_L^2 (1+z)^2} \sum_{a=-1}^1 \sum_{i=-1}^1 w_{a,i} \cdot E_{a,i} \cdot v_{a,i} \quad (14)$$

where the mode energies $E_{a,i}$ satisfy the modified dispersion relation:

$$E_{a,i}^2 = p^2 c^2 + \frac{\lambda_i p^n c^n}{M^{n-2}} + a\kappa(\hbar c)^2 \quad (15)$$

All three MDR terms are dimensionally consistent in units of J^2 . The group velocity is obtained by explicit differentiation of $E_{a,i}$ with respect to momentum, and the full geometric prefactor — luminosity distance, angular incidence, and cosmological redshift — appears transparently at the outermost level of the equation.

Validation against the solar constant and the isolated neutron star RX J1856.5–3754 confirms that the flat/luminal mode ($a = 0, i = 0$) yields flux predictions consistent with established observations. The quantum-sector structure of GPFF II provides a modular, physically transparent foundation for future investigations of Lorentz-invariance violation and curvature-dependent photon propagation in astrophysical settings.

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